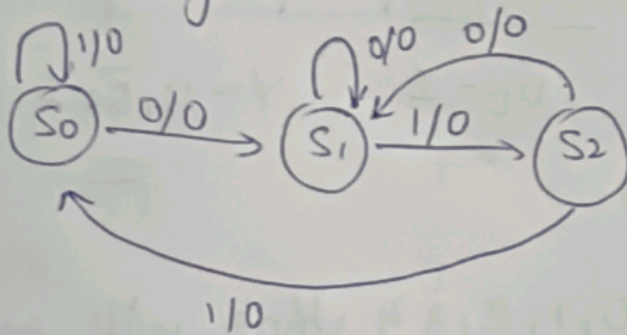


# FSM Assignment

- 1) Draw the state diagram for a FSM, the output toggles where -ver it detects '1110' LSB first (overlapping) consider value of output as '0'

→ State diagram



| Previous state | Input | next-state | Output Y |
|----------------|-------|------------|----------|
| 0              | 0     | 0          | 0        |
| 0              | 1     | 1          | 0        |
| 1              | 0     | 1          | 0        |
| 1              | 1     | 2          | 0        |
| 2              | 0     | 2          | 0        |
| 2              | 1     | 0          | 1        |
| X              | X     | X          | 0        |



| $D_1$ | $\overline{D_0}I$ |    |    |    |
|-------|-------------------|----|----|----|
|       | 00                | 01 | 11 | 10 |
| 0     | 0                 | 0  | 1  | 0  |
| 1     | 0                 | 0  | x  | x  |

$$D_1 = Q_0 \overline{I}$$

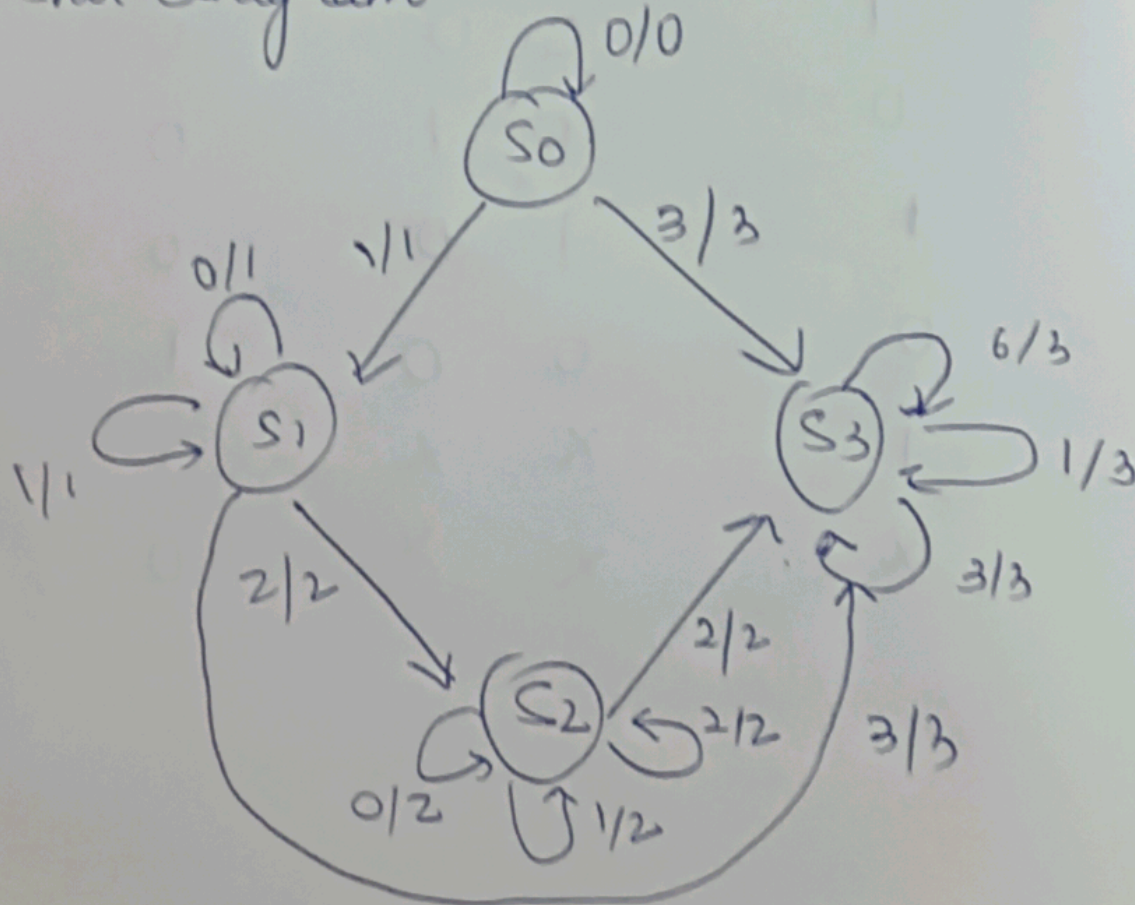
| $D_1$ | $\overline{D_0}I$ |    |    |    |
|-------|-------------------|----|----|----|
|       | 00                | 01 | 11 | 10 |
| 0     | 1                 |    |    |    |
| 1     | 1                 |    |    |    |

$$D_0 = I$$

$$Y = Q_1 \overline{D_0} I$$

② Design a FSM over  $\{0, 1, 2, 3\}$  that will output the largest input digit read so far eg input 011032 will produce output 001133.

+ State diagram



| Previous state | Input | next-state     |
|----------------|-------|----------------|
| S <sub>0</sub> | 0     | S <sub>0</sub> |
| S <sub>0</sub> | 1     | S <sub>1</sub> |
| S <sub>0</sub> | 2     | S <sub>2</sub> |
| S <sub>0</sub> | 3     | S <sub>3</sub> |
| S <sub>1</sub> | 0     | S <sub>0</sub> |
| S <sub>1</sub> | 1     | S <sub>1</sub> |
| S <sub>1</sub> | 2     | S <sub>2</sub> |
| S <sub>1</sub> | 3     | S <sub>3</sub> |
| S <sub>2</sub> | 0     | S <sub>0</sub> |
| S <sub>2</sub> | 1     | S <sub>1</sub> |
| S <sub>2</sub> | 2     | S <sub>2</sub> |
| S <sub>2</sub> | 3     | S <sub>3</sub> |
| S <sub>3</sub> | 0     | S <sub>0</sub> |
| S <sub>3</sub> | 1     | S <sub>1</sub> |
| S <sub>3</sub> | 2     | S <sub>2</sub> |
| S <sub>3</sub> | 3     | S <sub>3</sub> |

output